Project Design Phase-II
Technology Stack (Architecture & Stack)

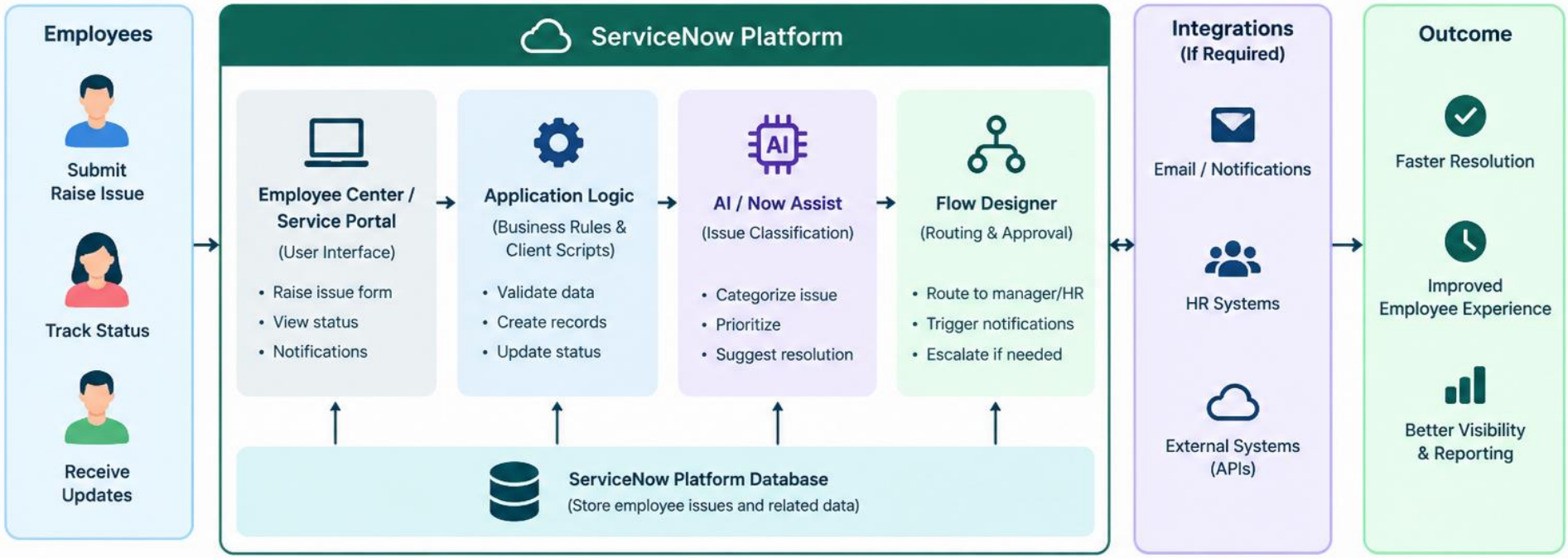
Date	12 September 2026
Project Name	Empolyee Raise Issue
Maximum Marks	4 Marks

Technical Architecture:

The Employee Raise Issue system is developed using **ServiceNow Service Portal and Record Producer**. Employees can submit issues through the portal, which creates a record in the Employee Issue table. **UI Policies, ACLs, roles, and Flow Designer** manage validation, security, assignment, notifications, and issue tracking.

Example: Employee issue reporting and tracking through ServiceNow Service Portal

Employee Raise Issue Process



Technology Stack

ServiceNow Employee Center / Service Portal	ServiceNow Business Rules & Client Scripts	ServiceNow Flow Designer	ServiceNow Platform Database	Now Assist / AI	REST API (If Required)	ServiceNow Cloud

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees use the portal to raise an issue, select Category and Subcategory, enter details, and track the issue status. .	ServiceNow Employee Center / Service Portal
2.	Application Logic-1	Controls the issue form and validates employee inputs while submitting an issue.	Client Scripts & UI Policies
3.	Application Logic-2	Displays relevant Subcategory choices based on the selected Category.	Dependent Choices / Dictionary Configuration
4.	Application Logic-3	Automatically assigns, routes, updates, and escalates employee issues based on defined conditions.	Business Rules & Flow Designer
5.	Database	Stores employee issue records, category, subcategory, description, priority, assignment, and status.	ServiceNow Tables / Platform Database
6.	Cloud Database	Provides cloud-based storage and access to employee issue data.	ServiceNow Cloud Platform
7.	File Storage	Stores attachments or supporting documents submitted along with employee issues.	ServiceNow Attachments
8.	External API-1	Sends notifications to employees when an issue is created, updated, assigned, or resolved	ServiceNow Email / Notification Service
9.	External API-2	Supports integration with external HR or organizational systems when required.	REST API / IntegrationHub
10.	Machine Learning Model	AI can be used to classify issues, suggest categories, prioritize issues, or assist with resolution.	ServiceNow AI / Now Assist
11.	Infrastructure (Server / Cloud)	Hosts the Employee Raise Issue application, database, workflows, and integrations.	ServiceNow Cloud Platform .

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No external open-source framework is required. The application is developed using ServiceNow's built-in platform features.	ServiceNow Platform, JavaScript
2.	Security Implementations	Controls employee access to issue records and restricts data based on user roles and permissions.	ServiceNow ACLs, Roles & Permissions, Authentication
3.	Scalable Architecture	The application is hosted on the ServiceNow cloud platform and can support multiple employees raising and tracking issues simultaneously.	ServiceNow Cloud Platform, Flow Designer
4.	Availability	The application is cloud-based and available to employees through the ServiceNow portal, allowing issues to be raised and tracked without maintaining local servers.	ServiceNow Cloud Infrastructure
5.	Performance	Client-side validation, dependent Category–Subcategory choices, automated workflows, and efficient database queries reduce unnecessary processing and improve response time.	Client Scripts, UI Policies, ServiceNow Tables, Flow Designer